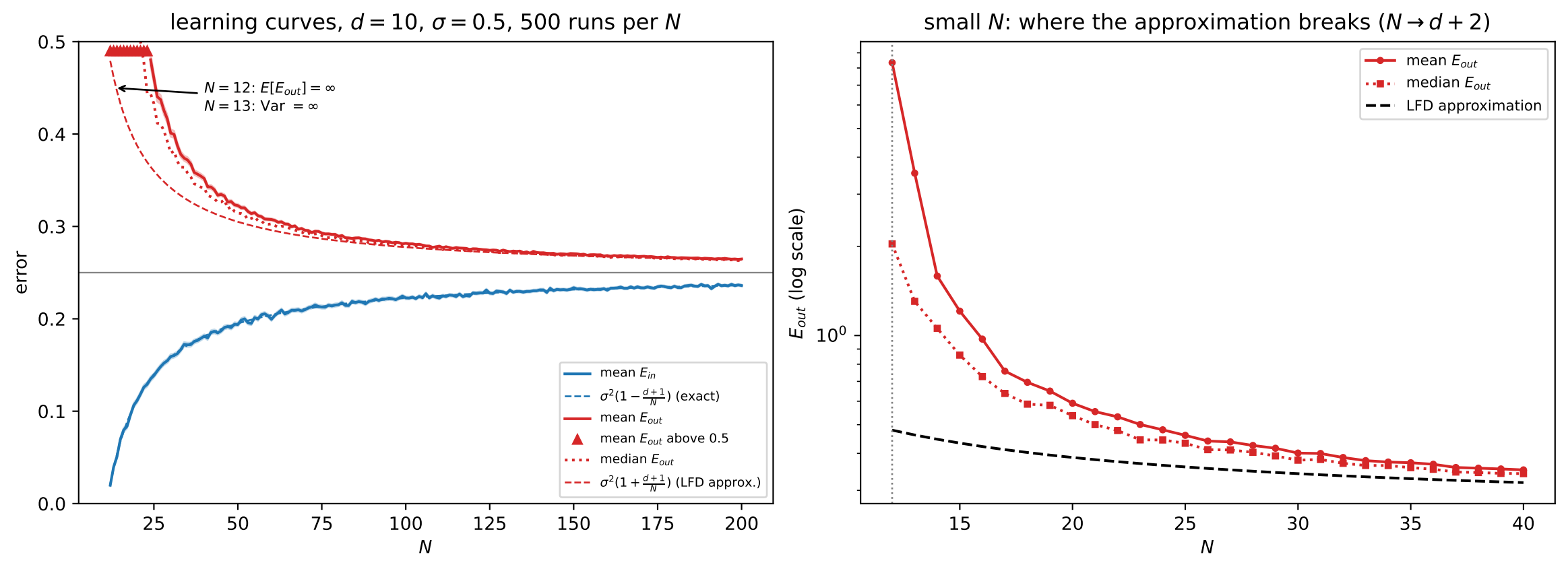
learning curves, $d = 10$, $\sigma = 0.5$, 500 runs per N
error
N
$N = 12$: $E[E_{out}] = \infty$
$N = 13$: Var $= \infty$
mean E_{in}
$\sigma^2(1 - \frac{d+1}{N})$ (exact)
mean E_{out}
mean E_{out} above 0.5
median E_{out}
$\sigma^2(1 + \frac{d+1}{N})$ (LFD approx.)
small N: where the approximation breaks ($N \to d + 2$)
E_{out} (log scale)
10^0
N
mean E_{out}
median E_{out}
LFD approximation